

STORM LAKE, IA, USA

WMO: 725496

Lat: 42.597N

Lon: 95.241W

Elev: 454

StdP: 95.99

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04976

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.4	-20.0	-27.1	0.3	-22.2	-24.1	0.5	-19.9	15.4	-9.1	14.2	-8.2	5.1	290	0.616

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	32.8	23.2	32.0	23.0	30.1	22.2	25.8	30.2	24.8	29.3	23.8	28.3	6.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.7	20.9	29.0	23.0	18.7	27.6	22.4	18.1	27.1	82.7	30.1	78.3	29.2	73.8	28.5	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.2	11.8	10.6	-26.9	35.8	3.4	1.5	-29.4	36.9	-31.4	37.8	-33.3	38.6	-35.7	39.7
			-27.0	27.4	4.2	1.0	-30.0	28.1	-32.4	28.7	-34.7	29.3	-37.8	30.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.2	-6.6	-5.0	2.1	9.3	16.0	21.6	23.3	21.6	18.1	10.7	2.7	-4.4
	DBStd	12.02	7.28	6.97	7.02	5.84	5.14	3.73	3.24	3.01	4.64	5.47	6.10	6.45
	HDD10.0	2021	514	421	261	80	9	0	0	0	2	58	228	448
	HDD18.3	3813	773	653	505	273	108	13	3	7	61	243	468	706
	CDD10.0	1722	0	1	14	61	195	348	412	359	244	79	10	0
	CDD18.3	474	0	0	1	4	35	111	157	107	53	6	0	0
	CDH23.3	4373	0	0	7	76	391	1076	1458	802	484	76	3	0
	CDH26.7	1486	0	0	1	22	136	401	530	225	152	19	0	0
Wind	WSAvg	5.1	5.7	5.7	5.8	6.3	5.9	4.8	3.7	3.3	4.0	5.0	5.5	5.5
Precipitation	PrecAvg	812	13	15	44	84	100	125	108	113	87	57	35	19
	PrecMax	1152	90	73	140	294	197	302	291	261	316	213	107	108
	PrecMin	353	0	0	0	5	21	13	14	6	18	8	0	0
	PrecStd	197	16	16	36	61	41	72	63	62	60	42	29	20
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.0	14.8	23.9	29.8	33.0	34.8	35.1	32.8	32.9	29.8	22.1	12.3
		MCWB	5.9	9.8	14.4	16.7	20.1	23.2	25.2	24.8	21.8	17.9	13.4	8.4
	2%	DB	7.2	10.0	19.7	25.2	30.1	32.6	32.6	31.1	30.3	25.2	17.5	8.7
		MCWB	4.0	7.1	12.7	14.8	18.6	22.2	24.7	24.7	21.4	16.2	11.4	5.8
	5%	DB	3.9	6.4	15.8	22.3	27.6	31.0	31.3	29.0	27.9	22.4	13.9	5.9
		MCWB	1.9	3.9	11.0	13.3	17.3	21.1	24.4	23.7	20.2	15.7	8.7	3.1
	10%	DB	2.3	3.7	12.3	19.0	25.0	28.9	29.9	27.5	26.3	19.8	11.3	3.0
		MCWB	1.0	1.9	8.5	11.6	16.3	20.3	23.8	22.3	19.0	13.8	7.5	1.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.4	10.3	16.5	18.2	22.5	26.1	27.6	26.9	24.9	21.4	15.1	9.9
		MCDB	11.2	13.7	22.0	24.0	29.3	31.1	31.9	30.9	28.8	26.1	19.2	11.0
	2%	WB	4.0	7.1	13.7	16.2	20.6	24.4	26.2	25.4	23.2	18.6	12.1	5.5
		MCDB	6.7	9.8	17.5	23.0	26.9	29.5	30.6	29.6	27.6	23.6	16.9	8.3
	5%	WB	2.3	4.1	11.1	14.5	19.1	23.2	25.2	24.2	22.1	16.4	9.8	3.6
		MCDB	3.8	6.5	15.6	20.5	25.1	28.4	29.8	28.0	26.5	20.3	13.4	5.5
	10%	WB	0.9	2.1	8.0	12.7	17.7	22.0	24.1	23.1	20.8	14.0	7.5	2.0
		MCDB	2.1	3.8	12.1	18.2	22.4	27.3	29.0	26.6	24.5	18.3	10.7	3.5
Mean Daily Temperature Range	5% DB	MDBR	8.6	8.9	9.9	11.7	11.5	11.0	10.4	10.0	12.3	11.2	10.1	8.2
		MCDBR	11.1	11.4	15.5	17.1	15.1	13.1	11.3	11.5	14.1	15.1	14.2	11.0
		MCWBR	8.7	8.7	9.3	8.8	6.6	5.6	6.2	6.7	6.8	7.9	8.6	8.4
	5% WB	MCDBR	10.2	10.8	14.5	14.4	12.5	11.0	10.2	9.8	11.4	12.4	13.0	9.8
		MCWBR	8.2	8.4	9.2	8.4	6.7	6.3	6.5	6.5	6.7	8.3	8.4	7.8
Clear-Sky Solar Irradiance	taub		0.278	0.290	0.316	0.365	0.390	0.406	0.416	0.407	0.370	0.330	0.291	0.277
	taud		2.415	2.399	2.421	2.311	2.311	2.322	2.273	2.295	2.364	2.461	2.499	2.452
	Ebn at Noon		871	917	930	902	884	868	853	850	859	858	851	841
	Edh at Noon		76	92	102	123	127	126	131	124	107	86	70	68
All-Sky Solar Radiation	RadAvg		1.91	2.96	3.81	4.80	5.64	6.26	6.55	5.51	4.47	3.05	2.08	1.56
	RadStd		0.18	0.23	0.35	0.42	0.41	0.50	0.35	0.38	0.28	0.38	0.13	0.11

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (54 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page